

$$\begin{bmatrix} 2 \\ -4 \\ 2 \\ -6 \end{bmatrix} \div 2 \quad \begin{bmatrix} 3 \\ -9 \\ 12 \end{bmatrix} \div 3 \quad \begin{bmatrix} 2 \\ 4 \end{bmatrix} \div 2 \quad \begin{bmatrix} 5 \end{bmatrix} \div 5 \rightarrow L = \begin{bmatrix} 1 & 0 & 0 & 0 \\ -2 & 1 & 0 & 0 \\ 1 & -3 & 1 & 0 \\ -3 & 4 & 2 & 1 \end{bmatrix}$$

An easy calculation verifies that this L and U satisfy $LU=A$ $\square$

In practical work, row interchanges are nearly always needed, because partial pivoting is used for high accuracy.

To handle row interchanges, the LU factorization above can be modified easily to produce an L that is permuted lower triangular, in the sense that a rearrangement (called a permutation) of the rows of L can make L (unit) lower triangular.

The resulting permuted LU factorization solves $Ax=b$ in the same way as before, except that the reduction of $[Lb]$ to $[Iy]$ follows the order of the pivots in L from left to right, starting with the pivot in the first column.

PRACTICE PROBLEM

Find an LU factorization of $A = \begin{bmatrix} 2 & -4 & -2 & 3 \\ 6 & -9 & -5 & 8 \\ 2 & -7 & -3 & 9 \\ 4 & -2 & -2 & -1 \\ -6 & 3 & 3 & 4 \end{bmatrix}$

Solution:

$$A = \begin{bmatrix} 2 & -4 & -2 & 3 \\ 6 & -9 & -5 & 8 \\ 2 & -7 & -3 & 9 \\ 4 & -2 & -2 & -1 \\ -6 & 3 & 3 & 4 \end{bmatrix}$$

$\xrightarrow[\text{Row 1}]{\text{Row 3} = \text{Row 3} - \text{Row 1}}$

$$\begin{bmatrix} 2 & -4 & -2 & 3 \\ 0 & 3 & 1 & -1 \\ 0 & -3 & -1 & 6 \\ 4 & -2 & -2 & -1 \\ -6 & 3 & 3 & 4 \end{bmatrix}$$

$\xrightarrow[\text{+ } 3 \times \text{Row 1}]{\text{Row 5} = \text{Row 5} - \text{Row 1}}$

$$\begin{bmatrix} 2 & -4 & -2 & 3 \\ 0 & 3 & 1 & -1 \\ 0 & -3 & -1 & 6 \\ 0 & 6 & 2 & -7 \\ 0 & -9 & -3 & 13 \end{bmatrix}$$

$\xrightarrow[\text{3} \times \text{Row 1}]{\text{Row 2} = \text{Row 2} - \text{Row 1}}$

$\xrightarrow[\text{-2} \times \text{Row 1}]{\text{Row 4} = \text{Row 4} - \text{Row 1}}$

$\xrightarrow[\text{+ Row 2}]{\text{Row 3} = \text{Row 3} - \text{Row 2}}$

A_1

$$\begin{bmatrix} 2 & -4 & -2 & 3 \\ 0 & 3 & 1 & -1 \\ 2 & -7 & -3 & 9 \\ 4 & -2 & -2 & -1 \\ -6 & 3 & 3 & 4 \end{bmatrix}$$

$$\begin{bmatrix} 2 & -4 & -2 & 3 \\ 0 & 3 & 1 & -1 \\ 0 & -3 & -1 & 6 \\ 0 & 6 & 2 & -7 \\ -6 & 3 & 3 & 4 \end{bmatrix}$$

$$\begin{bmatrix} 2 & -4 & -2 & 3 \\ 0 & 3 & 1 & -1 \\ 0 & 0 & 0 & 5 \\ 0 & 6 & 2 & -7 \\ 0 & -9 & -3 & 13 \end{bmatrix}$$